Universität Konstanz

Abstraction Mechanisms for Virtual Avatars

Master's Thesis Defense

Nidanur Günay

First reviewer: Prof. Dr. Oliver Deussen
Second reviewer: Prof. Dr. Tiare Feuchtner

University of Konstanz, 2026

Motivation and Problem Statement

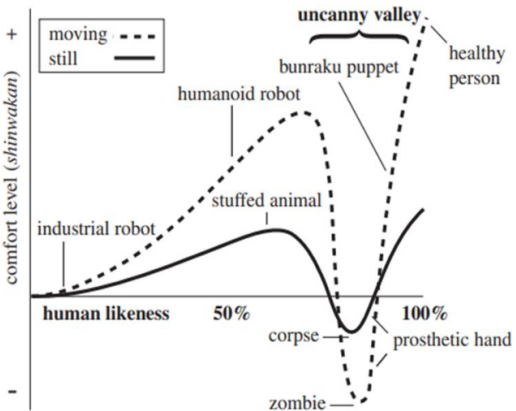
Chapter 1

The Setting

- CG humanoid characters present, teach, advise and sell
- So how they are **drawn** decides how much we believe them

The Problem

- Mori 1970: comfort rises with human likeness, **but only to a point**. Close to human but not quite is the **uncanny valley** [1]
- Proposed for robots, later confirmed for **CG characters**
- A photorealistic avatar may undermine the credibility it was built for

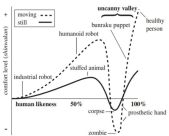


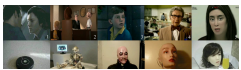
Mori 1970: comfort against human likeness [1]

Related Work

1. Measuring the Valley

Chapter 3 — from a drawn curve, to data, to an instrument, to CG characters





01

1970, Mori

The Uncanny Valley [1]

Proposed it. Comfort drops just short of human, and **motion deepens the dip**.

02

2006, MacDorman & Ishiguro

The Uncanny Advantage of Using Androids [2]

Measured it. Eeriness peaks where **comfort is lowest**.

03

2010, Ho & MacDorman

Revisiting the Uncanny Valley Theory [3]

Instrumented it. Eeriness becomes **separable from likeness**.

04

2012, Dill et al.

Evaluation of the Uncanny Valley in CG Characters [4]

Extended it. It appears for **CG characters**, worst at the **eyes and mouth**.

Takeaway. The valley is real, measurable and reaches CG avatars. None of it says whether a viewer would *rely* on one.

2. Trust and the Valley

Chapter 3 — the valley reaches trust, and trust has three parts

01

2005, Nowak & Rauh

The Influence of the Avatar on Online Perceptions [5]

Tested credibility. More human-like avatars rated **more credible**.

So How Is Trust Measured?

Ability. Can they actually do this? Competence inside one domain, not in general.

Benevolence. Are they on my side? The belief that the other party cares about my wellbeing.

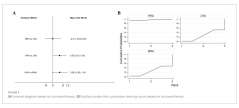
Integrity. Do they play straight? Principles I accept, and saying what they mean.

Mayer et al. 1995: trustworthiness has three *separable* dimensions, not one score [6]. All three measured, once per advisor.

Takeaway. The valley reaches trust, not only comfort. And trust is not one number, which is why nothing in this talk reports a single trust score.

3. Abstraction as the Exit

Chapter 3 — three studies, one meta-analysis, and they do not agree



01

2024, Canales et al.

The Impact of Avatar Stylization on Trust [7]

Same Mayer model. Three levels. Self-report **flat**, but 48% chose the **semi-realistic** doctor.

02

2025, Dubosc et al.

Effect of Avatar Stylization and Facial Expression Intensity [8]

Face to face in VR. Stylised avatars rated significantly **less eerie**.

03

2025, Gao et al.

Trust in Virtual Agents [9]

An agent against the real speaker. The **real speaker** won on all three dimensions, but **not on overall trust**.

04

2025, Tao et al.

Impact of Virtual Avatar Appearance Realism [10]

Meta-analysis, 13 studies. **High realism** more trustworthy than low, but not than medium.

Takeaway. Two say abstraction helps, two favour realism, and neither side is clean. The evidence has not converged, which is what RQ3 goes after.

Research Questions

Chapter 1 — three gaps in the work just shown, three questions

Three Research Questions

RQ1 Technical transfer — **System**

How can NPR techniques be adapted to platforms with different rendering architectures, and what capabilities and limitations does each impose?

RQ2 Technique selection — **Experiment 1**

Which technique or combination best balances visual abstraction against facial and expressive legibility?

RQ3 Perception — **Experiment 2**

How does stylisation affect eeriness and trustworthiness, and can it reach the trustworthiness of a photorealistic avatar and of a real human speaker?

Four Contributions

- **System:** eight NPR techniques, three production avatar platforms, runtime-selectable
- **Calibration:** parameters set by preference testing, not by eye
- **Design:** ratings and behavioural choice together, rendering style isolated from confounds
- **Evidence:** trust against a photorealistic avatar *and* a real human

Methodology and Evaluation

Three Avatar Platforms

Chapter 4.2 — chosen to span the range of pipeline access

	Mixamo	Meta SDK	Avaturn
Shader access	**Full URP**, unrestricted	**Hook injection only**, after SDK lighting	**Full URP**, unrestricted
Facial animation	**None**	Requires **Quest Pro** hardware	**Blendshape animation**
Key challenge	No facial rigging	Shader manipulation	Initial Unity import
Strength	Full shader freedom [11]	Native Quest 3 integration [12]	**High fidelity facial quality** [13]

Why Avaturn for the Study

- The only one with a **realistic face** that can also be **animated**
- The study avatar was built from photographs of the real speaker
- All three were used to develop the eight techniques

Eight NPR Techniques

Chapters 4 and 6.1 — built iteratively, then assessed against four criteria

Technique	What it adds	Facial readability	Temporal stability	Artefact suppression	Expressive abstraction	Candidate
V1 Geometric Silhouette Outline	Silhouette from geometry	✓	✓	(✓) alpha cutout	× not in isolation	✓
V2 Toon Shading	Quantised diffuse bands	✓	✓	✓	(✓) no interior lines	—
V2.2 X-Toon Shading	2D ramp: lighting *and* abstraction	✓	✓	✓	(✓) manual mode	✓
V3 Screen-Space Normal Edge Detection	Creases from normal derivatives	× landmarks missed	✓	✓	✓	—
V4 Sobel Edge Detection	Colour-buffer gradients	(✓) threshold	(✓)	(✓) preprocessing	✓	✓
V4.2 Gaussian Prefiltered Sobel	Smoothing before differentiation	× no gain over V4	(✓)	✓	✓	—
V5 Hierarchical Edge Detection with Multiple Cues	Depth, normal and colour merged	× segments unclear	✓	× nose bridge	✓	✓
V6 Kuwahara Painterly Filter	Painterly surface abstraction	× at large radius	✓	✓	✓	✓

✓ satisfactory without tuning (✓) satisfactory after tuning × limitation observed

Each technique was built to resolve a limitation of the one before. **No technique satisfied all four criteria alone**, and **facial readability** failed most often, four times out of eight. **Five went forward** as candidates for the composite.

Applying the Techniques

Chapter 4 — the texture based detectors failed on the study avatar

V4.2 Gaussian Prefiltered Sobel

V5 Hierarchical Edge Detection with Multiple Cues

What the Detectors Produced

- Lines where there is no feature
- **Eyebrows and lip line barely register**
- V4.2: smoother, features still missing
- V5: brow and nose shadow become hard edges

Why

- Head rebuilt from **photographs**
- **Baked illumination** and **fine skin noise** both read as structure
- Smoothing removes the noise *and* the real edges together
- No threshold separates them

Avatar Texture Manipulation

Chapter 4 — fixing the input instead of the operator

Three Edits in Blender

- Baked illumination blurred out
- Skin tone unified
- Mouth saturation raised, lip line added

What It Recovered

- **Lip line and eyebrows** now detectable
- Gaussian prefilter no longer needed

Scope

- Applied to the **NPR avatar** only
- Photorealistic version keeps original texture

Default Avaturn

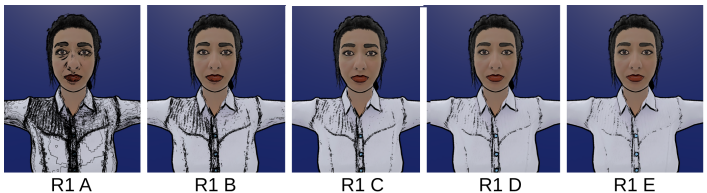
Preprocessed head texture

Experiment 1 — Parameter Adjustment Study

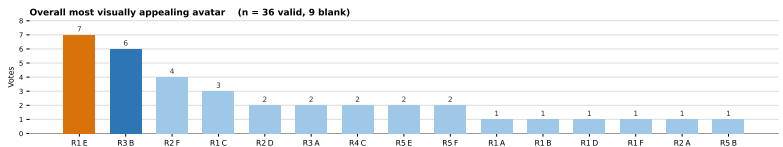
Parameter Adjustment Study — Design

Chapter 4.8 — calibration by preference, not by inspection

- **45 participants** ranked rendered variants of the avatar by **perceived trustworthiness**
- One parameter group per round, five rounds, and each round's outcome frozen for every round after it
- A closing question asked for the **overall most visually appealing** avatar across all rounds



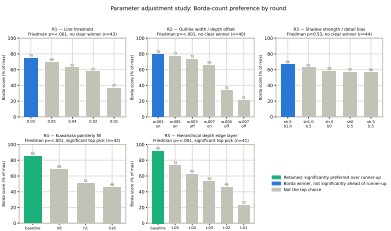
One round as an example. R1 inner line threshold, 0.01 to 0.05.



Parameter Adjustment Study — Outcome

Chapter 6.2.1 — what each round settled

	Parameter group	Winner	Significant?	Set to
R1	Inner line threshold	0.03	no, $p = .31$	**0.05** appeal
R2	Outline width, depth offset	0.003, on	no, $p = .82$	**0.003, on** Borda
R3	Shadow strength, detail bias	0.5 / 1.0	no, $p = .53$	**0.0 / 0.5** appeal
R4	Kuwahara hardness, sharpness	**none**	**yes,** $p = .003$	**off** baseline won
R5	Hierarchical depth threshold	**none**	**yes,** $p < .001$	**off** baseline won



Borda score per round. Green = significantly preferred, blue = top choice, not clearly ahead.

- Ranks are **ordinal**, so the tests are non-parametric
- **Borda** scores the whole ballot, **Friedman** tests the round against chance, **Wilcoxon** compares the top two
- **Kuwahara** and the **hierarchical layer** dropped
- Composite: **outline, X-Toon, Sobel**

RQ1 and RQ2 Answers

Chapters 4 and 6 — what the system work established

RQ1 Adaptation across Architectures

- All eight techniques ran on **all three platforms**
- Same values, visibly different output per host: **retuning always required**
- **Meta SDK**: one fragment hook, no second pass, no depth buffer
- V5's depth cue became a **camera-distance derivative**; V6's structure tensor lost its **smoothing pass**

RQ2 Abstraction versus Legibility

- On the raw texture, **texture-based edge detection failed**
- After preprocessing, Sobel finds the **eyebrows and lip line**; V4.2 no longer repays its 72 samples
- **Five of the eight** went forward as candidates
- The parameter study excluded **V5** and **V6**
- The answer is a **composite**: V1 Geometric Silhouette Outline, V2.2 X-Toon Shading and V4 Sobel Edge Detection, over a preprocessed head texture

Experiment 2 — Main User Study

Main User Study — Design

Chapter 4.10 — within-subjects, video based, three conditions

Design

- 3 conditions × 3 scenarios (career, travel, learning)
- Three **everyday domains**, avoiding carryover
- Recommended option alternates **A, B, A**
- Paradigm after Canales et al. 2024 [7]

Order — 3 × 3 Latin Square, each condition once per slot

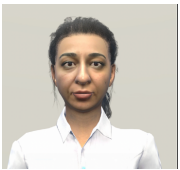
Version	Video 1 Career	Video 2 Travel	Video 3 Learning
A	C2	C1	**C3**
B	**C3**	C2	C1
C	C1	**C3**	C2

Why Video Rather than VR

- C1 has **no VR equivalent**
- Video fixes **one recorded performance**: the C2/C3 blendshape stream was captured while C1 was filmed

C1 — Real

C2 — Photorealistic



C3 — NPR

Same person, same animation, same voice. **C2 vs C3** is the critical pair.

Study Procedure

Decision vignette → Baseline choice → Advisor video 23 s → Questionnaire Parts A to G

one block, repeated × **3**, once per condition

Three forced rankings — all three advisors ordered
most want to consult · most trustworthy · least unsettling

Hypotheses and the Subscales

Chapter 4.10.1 — what was predicted, and what the ratings did

4 Hypotheses & Statistical Tests

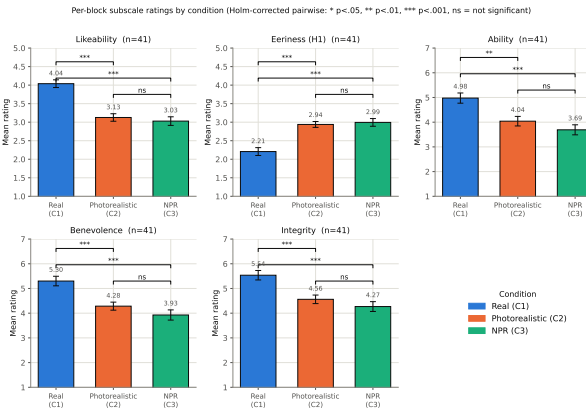
H1 NPR (C3) is rated *less eerie* than the photo-realistic avatar (C2).
Test: RM-ANOVA, Holm pairwise

H2 NPR (C3) does *not* cost trust relative to C2.
Test: Pairwise plus TOST equivalence, $d = 0.3$ bound

H3 Eeriness lowers trust, likeability raises it, within participant.
Test: Person-centred multilevel regression

H4 NPR (C3) does *not* cost trust relative to the real speaker (C1).
Test: Same as H2, against C1

H2 and H4 predict a **null**. A non-significant p is not equivalence, hence the **TOST** bound, fixed in advance.



The real speaker separates from both avatars on every subscale. The two avatars separate from each other on *none* of them.

Measures and Analysis Plan

Chapter 4.10.4 and 6.2.4 — validated instruments, the power carried, and did they see the difference?

Per-Block Questionnaire

A	Comprehension, gating	**exclusion rule**
B	Likeability, 5 items	Godspeed, Bartneck et al. 2009 [14]
C	Eeriness, 8 items	Ho & MacDorman 2010 [3]
D	Ability, 3 items	Mayer et al. 1995 [6], wording adapted from Canales et al. 2024 [7]
E	Benevolence, 3 items	Mayer et al. 1995 [6], wording adapted from Canales et al. 2024 [7]
F	Integrity, 3 items	Mayer et al. 1995 [6], wording adapted from Canales et al. 2024 [7]
G	Recommendation agreement	secondary

B and C on a 5-point semantic differential, D to G on a 7-point Likert.
Composite reliability $\alpha = .87$ to $.96$.

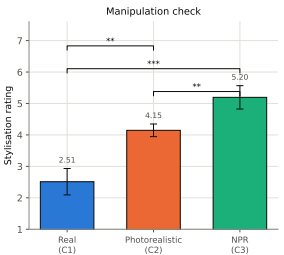
Analysis Plan

- RM-ANOVA, Holm-corrected pairwise
- **TOST equivalence**, $d = 0.3$ bound, for H2 and H4
- Person-centred multilevel regression for H3
- Friedman and Wilcoxon for the rankings

Sample Size

- Power target $n = 38$, recruited for **48**
- 52 enrolled, **11 excluded** on gating
- $n = 41$ valid in all three, age $M = 27.6$
- A paired TOST needs near $n = 120$

Did They See the Difference?



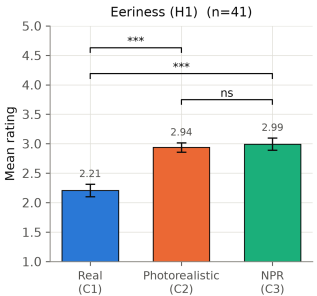
How stylised each advisor looked, 1 to 7.

Omnibus	$F(2, 80) = 15.45, p < .001, \eta_p^2 = .28$
C1 vs C2	$p = .001, g = 0.87$
C1 vs C3	$p < .001, g = 1.23$
C2 vs C3	$p = .004, g = 0.55$

All three conditions separate.

Results

H1 — Eeriness, C2 vs C3



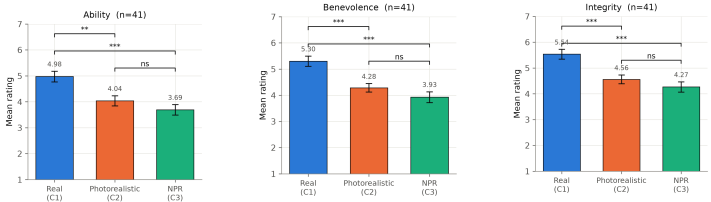
H1 NOT SUPPORTED

Eeriness	C1 vs C2	C1 vs C3	C2 vs C3
p	$< .001$	$< .001$	.66
g	0.98	0.98	0.07

Omnibus: $F(2, 80) = 20.39$, $p < .001$, $\eta_p^2 = .34$

- Eeriness rose from C1 to C2 to C3, with C3 *above* C2, the **opposite** of the prediction
- Real speaker differs from both avatars
- The two avatars **statistically indistinguishable**, $g = 0.07$
- **Stylisation and eeriness are separate effects**: seen, but not felt as less eerie

H2 — Trust, C2 vs C3



ns between the two avatars on all three trust dimensions.

H2 SUPPORTED

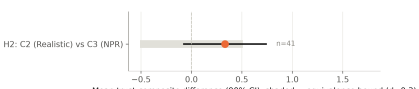
- **No statistically significant difference** across the trust subscales
- **No difference detected**, but equivalence unestablished
- Realised power near 0.20 at $n = 41$
- Roughly **three times** the sample required to separate the conditions

Step 1. Pairwise, C2 vs C3

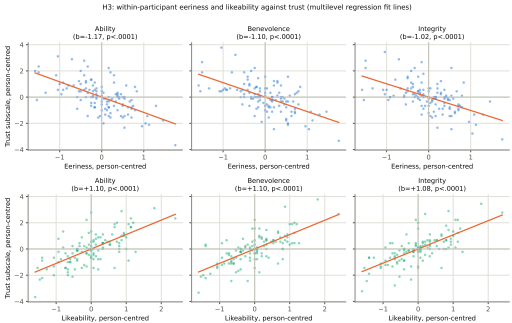
	p	*g*
Ability	.21	0.20
Benevolence	.18	0.19
Integrity	.27	0.16

Step 2. TOST, trust composite

	Diff.	Bound	TOST *p*
C2 vs C3	0.33	±0.51	.240



H3 — Eeriness, likeability and trust



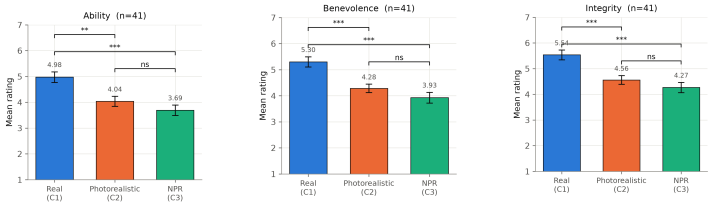
H3 CONFIRMED

- **Within** participants, not between conditions
- Eeriness **lowers** trust, likeability **raises** it
- About **one point for one point**, on every dimension
- Confirmed in **both** directions
- Caveat: the two correlate at $r = -.72$, and cannot be separated cleanly here

Predictor	Eeriness	Likeability
Ability	-1.17	$+1.10$
Benevolence	-1.10	$+1.10$
Integrity	-1.02	$+1.08$

Fixed-effect slopes, all $p < .001$

H4 — Trust, C1 vs C3



The real speaker is **significantly above** the NPR avatar on all three.

H4 NOT SUPPORTED

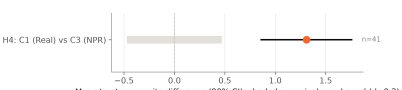
- All three subscales **significantly higher** for the real speaker
- Nearly **three times** the equivalence bound
- Not a marginal shortfall
- Two explanations stay open: the uncanny valley, or participants recognising the speaker

Step 1. Pairwise, C1 vs C3

	p	*g*
Ability	$< .001$	0.75
Benevolence	$< .001$	0.79
Integrity	$< .001$	0.80

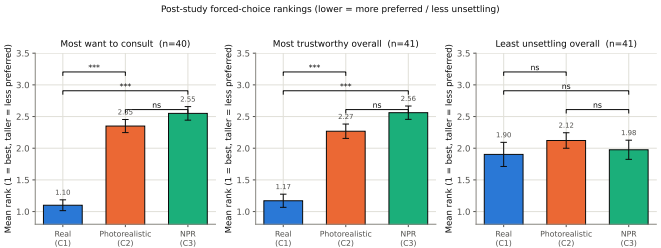
Step 2. TOST, trust composite

	Diff.	Bound	TOST *p*
C1 vs C3	1.31	± 0.47	.998



The Behavioural Measure

Chapter 6.2.6 — the three rankings



Question	Friedman	C1 vs C2/C3	C2 vs C3
Most want to consult	$\chi^2_F(2) = 49.40, p < .001$	$p < .001$	$p = .21$
Most trustworthy	$\chi^2_F(2) = 44.05, p < .001$	$p < .001$	$p = .07$
Least unsettling	$\chi^2_F(2) = 1.02, p = .60$	$p = .97$	$p = .97$

Lower is better — rank 1 = most preferred.

The real speaker ranks first. The **two avatars tie**.

RQ3 Answer

What stylisation did to eeriness and trustworthiness, and how far it reached

What Did Not Happen

- Eeriness **did not fall**, it rose slightly
- The two avatars **indistinguishable** on all five subscales
- No behavioural advantage either

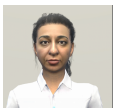
What Did Happen

- **No trust penalty** detected for stylisation
- Within a person, eeriness **lowers** trust and likeability **raises** it
- About **one point for one point**, on all three trust dimensions
- The route from appearance to trust is real, the composite just never used it

The Dividing Line

- The gap is **not between render styles**
- C2 vs C3: 0.33 points, inside the noise
- C1 vs C3: 1.31 **points**, nearly three times the bound

C1 — Real



C2 — Photorealistic

C3 — NPR

Discussion

Why the two avatars scored the same

Two mismatches inside the composite

What Needs Explaining

- Real speaker **far above** both avatars
- The two avatars: **seen as different**, **scored the same**
- H3: appearance **moves trust**, but the composite never used that route

1. Material vs. Shape

- The composite stylises the **material**, not the **shape**
- Zell et al. 2015: eeriness **rises** when material and shape disagree [15]
- Stronger stylisation would only **widen** the gap

2. Appearance vs. Performance

- Both avatars share the **same** recorded voice and animation
- C3 puts a **human performance** on an abstracted face
- Cihodaru 2025, Kätsyri et al. 2015: mismatched human likeness is **rated eerier** [16, 17]

What this design could not show

Chapter 6.3 — four factors this design cannot rule out

1. The Comparison Set

- Real speaker sets the top: only **how far below?**
- Rendering gap vs. **human versus CG** gap
- Fits the tiny C2–C3 likeability and eeriness gap
- C3 right after C1 in 2 of 3 versions, C2 **never**
- Without C1, abstraction is the **only axis**

2. Familiarity with the Speaker

- C1 is **the researcher**
- Recognition adds trust **no rendering can match**
- Lifts C1 above a stranger, widening the gap
- No recognition item: share **unknown**

3. The Video Setting

- Flat, prerecorded, **no interaction**
- Zibrek & McDonnell 2019: in VR, realism raised **presence** [18]
- Interaction lets mismatches show more
- VR **may separate** C2 and C3

4. Expectations of Generative Video

- One participant asked **which AI** made the videos
- One remark: an **observation**, not a result
- AI doubt lowers **both** avatars equally
- Nobody was told the videos were **not** AI

Limitations

Five, and each one bounds a different claim

Session Length

- 10 to 15 minutes per study
- Every measure completed **three times**
- Capped the measures and the breadth of the design

One Configuration

- One composite, one avatar, one platform, three scenarios
- Characterises **this configuration**, not NPR as a category

Parameter Study Scope

- Only the **most influential** parameters tested
- The rest fixed by inspection in development
- Highest **among those tested**, not necessarily the best available

Sample Size

- Deadline set the collection window, in the semester break
- $n = 41$ **detects** a difference
- **Not** the far larger sample equivalence demands

Counterbalancing

- **Full** balancing of three conditions needs all **six** orders
- Six would have demanded about **twice** the sample
- A Latin square with **three** was used instead

Conclusion

What was built, what it showed, and what it means

What This Thesis Built

- NPR stylisation for **Mixamo, the Meta SDK and Avaturn**
- Eight techniques in sequence, each answering a limitation of the one before
- Narrowed to **three** by visual assessment and a parameter adjustment study

What the Study Showed

- The composite **did not reduce eeriness** against the photorealistic baseline
- It produced **no observed trust penalty** either
- The evidence does not indicate stylisation reduced trust
- It is **not sufficient** to establish that the two are equivalent

What It Means

- A developer choosing between the two styles need not assume **an inherent disadvantage**
- The gap that matters is **not between render styles**
- It is the distance to the **real human**, and neither avatar closed any part of it

Reaching the credibility of an actual human is a harder challenge than selecting a rendering style.
It is also the problem that decides whether virtual advisors become genuinely trusted participants rather than a passing novelty.

Future work

Six steps, in the order the thesis sets them out

1. Larger Replication	2. Stylise the Geometry	3. Drop the Real Speaker	4. Rule Out Generative AI	5. Sweep Abstraction	6. Restore Interaction
– Repeat C2 versus C3 with roughly **120 participants** – Settles the equivalence question either way	– Shift proportion and feature scale **alongside the shading** – Tests the material-versus-shape mismatch – Match **voice and motion** to the stylised face	– Removes the human **anchor** – Shows whether the two avatars separate with no human in view	– State at the outset that the videos were **not AI-generated** – That belief may have lowered trust for both avatars	– Replace the single point with a **curve** – Can reveal an optimum between full realism and full abstraction	– Reciprocal behaviour and shared space, in **VR** – Across multiple avatar identities and platforms

Thank you

Questions

Eight NPR techniques, three avatar platforms, two studies, 86 participants.

Stylisation was clearly seen, cost no measurable trust,
and did not close the gap to a real human.

Nidanur Günay | nidanur.guenay@uni-konstanz.de

References

[1] Mori, M. The Uncanny Valley. *Energy* 7(4), 33–35, 1970.

[2] MacDorman, K. F., Ishiguro, H. The Uncanny Advantage of Using Androids in Cognitive and Social Science Research. *Interaction Studies* 7(3), 297–337, 2006.

[3] Ho, C.-C., MacDorman, K. F. Revisiting the Uncanny Valley Theory. *Computers in Human Behavior* 26(6), 1508–1518, 2010.

[4] Dill, V. et al. Evaluation of the Uncanny Valley in CG Characters. *IVA 2012*, 511–513.

[5] Nowak, K. L., Rauh, C. The Influence of the Avatar on Online Perceptions of Anthropomorphism, Androgyny, Credibility, Homophily, and Attraction. *JCMC* 11(1), 153–178, 2005.

[6] Mayer, R. C., Davis, J. H., Schoorman, F. D. An Integrative Model of Organizational Trust. *AMR* 20(3), 709–734, 1995.

[7] Canales, R., Roble, D., Neff, M. The Impact of Avatar Stylization on Trust. *IEEE VR*, 418–428, 2024.

[8] Dubosc, C. et al. Effect of Avatar Stylization and Facial Expression Intensity in Virtual Interactions. *Virtual Reality* 29(4), 2025.

[9] Gao, Y. et al. Trust in Virtual Agents: Exploring the Role of Stylization and Voice. *IEEE TVCG* 31(5), 3623–3633, 2025.

[10] Tao, Z. et al. Impact of Virtual Avatar Appearance Realism. *Frontiers in Psychology* 16, 2025.

[11] Adobe Inc. Mixamo: 3D Character Rigging and Animation Service. `https://www.mixamo.com/`, 2024.

[12] Meta Platforms, Inc. Meta Avatars SDK Documentation. `https://developers.meta.com/horizon/documentation/unity/meta-avatars-overview/`, 2024.

[13] Avaturn. Avaturn: Realistic Avatars from a Photo. `https://avaturn.me/`, 2024.

[14] Bartneck, C. et al. Measurement Instruments for Anthropomorphism, Animacy, Likeability... *Int. J. Social Robotics* 1(1), 71–81, 2009.

[15] Zell, E. et al. To Stylize or Not to Stylize? *ACM TOG* 34(6), 184:1–184:12, 2015.

[16] Cihodaru-Ştefanache, Ş., Podina, I. R. The Uncanny Valley Effect in Embodied Conversational Agents. *Frontiers in Psychology* 16, 2025.

[17] Kätsyri, J. et al. A Review of Empirical Evidence on Different Uncanny Valley Hypotheses. *Frontiers in Psychology* 6, 390, 2015.

[18] Zibrek, K., McDonnell, R. Social Presence and Place Illusion are Affected by Photorealism in Embodied VR. *ACM MIG*, 1–7, 2019.

[19] McDonnell, R., Breidt, M., Bülthoff, H. H. Render Me Real? *ACM TOG* 31(4), 91:1–91:11, 2012.

[20] Seymour, M. et al. Have We Crossed the Uncanny Valley? *JAIS* 22(3), 591–617, 2021.

[21] Alipour, A., Hartmann, T., Alimardani, M. Would You Rely on an Eerie Agent? *Computers in Human Behavior Reports* 23, 2026.

[22] Latoschik, M. E. et al. The Effect of Avatar Realism in Immersive Social Virtual Realities. *ACM VRST*, 2017.

[23] Lakens, D. Equivalence Tests: A Practical Primer for *t* Tests, Correlations, and Meta-Analyses. *Social Psychological and Personality Science* 8(4), 355–362, 2017.

[24] Demšar, J. Statistical Comparisons of Classifiers over Multiple Data Sets. *JMLR* 7, 1–30, 2006.

Media credits

Video, slide 2

Mark Zuckerberg: First Interview in the Metaverse, Lex Fridman Podcast #398. YouTube.
youtube.com/watch?v=MVYrJJNdrEg

Avaturn: Realistic 3D Avatars for Gaming from Photo. YouTube.
youtube.com/watch?v=rSb0Io92_R8

How Apple Vision Pro's New Spatial Persona Changes Everything. YouTube, 11 April 2024.
youtube.com/watch?v=H3PB2zA6Q-c

Excerpts shown for scholarly illustration under the German right of quotation, § 51 UrhG.

Figures

Slide 2, uncanny valley curve: Mori, reference [1].

Slide 4, android photograph: reference [2].

Slide 5, avatar set: reference [5].

Slides 4 to 6: every figure is reproduced from the study cited in the same card. The number under the title covers the figure as well as the claim, references [1] to [10].

Slides 10, 11, 13, 14, 16, 17 and 19, and all backup slides: own work. Renders from the Unity project, plots from the study data.

Backup, design compared with Zibrek: the render style images and the result chart are reproduced from reference [18].

Avatars: Avaturn, Meta Avatars SDK and Mixamo, used under their respective terms.

Logo: University of Konstanz.

Backup slides

Backup — How the evidence developed

Fifty years of it, and it still points both ways

Period	What changed	The evidence	Points towards
1970 to 2012	The valley is real, and it reaches avatars	Mori proposes it [1], MacDorman & Ishiguro measure it [2], Dill et al. confirm it for CG film and game characters [4]	valley confirmed
2012	Render style moves *trust*, in the same shape	McDonnell et al.: eleven styles on two actors. Appeal, reassurance and trustworthiness all valley-shaped, and the cartoon styles sat in the top trust group [19]	**abstraction**
2015	What drives it is the material, and consistency	Zell et al.: material governs eeriness, and a mismatch between material and shape makes it worse [15]	depends on the pairing
2024 to 2025	Abstraction looks promising	Canales et al.: self-report equal, semi-realistic won the behavioural choice [7]. Dubosc et al.: stylised significantly less eerie [8]	**abstraction**
2021 to 2026	But realism keeps winning too	Seymour et al.: photorealistic beat a caricature on trust [20]. Gao et al.: real speaker higher on all three dimensions, not on overall trust [9]. Tao et al.: high realism more trustworthy than low [10]. Alipour et al.: inconsistent across 53 studies [21]	**realism**

The last two rows are the reason this thesis exists. The evidence has not converged, and nobody has tested one avatar under both treatments against the real person it was built from.

Backup — Platform integration

Technical review 1/6: how the NPR shader reaches the avatar on each platform

	Mixamo	Avaturn	Meta SDK
Asset	FBX, standard skinned mesh	`.glb` (glTF 2.0), loaded at runtime by **GLTFast**, blendshapes kept	Built by the SDK at runtime, shader generated from templates
Applying NPR	Direct **material replacement**: custom multi-pass NPR shader per material (Avaturn after load)		NPR shader named in the SDK's **shader configuration asset**, code injected in `AppSpecificPostManipulation`
Switching	Replace the active material set		One shader, technique chosen by an **active keyword**
Materials	4 submesh materials	6 submesh materials (skin, hair, eyelashes . . .)	Managed by the SDK
Inputs	Raw albedo, normal map, lighting term, URP **depth, normal and colour buffers**		Lit colour, interpolated normal, world position. **No depth-normals pass**, posed mesh only inside the SDK shader
Extra passes	**Yes**: URP renderer feature, avatar mask on its own rendering layer		**No**: a mask pass would draw the bind pose
Normal map	Unity importer, `UnpackNormalScale`	`rgb*2-1`, GLTFast skips the importer	Decoded by the SDK before the hook
Face	**No facial rig**	52 ARKit blendshapes, Live Link Face	Requires **Quest Pro**

Unity 2022.3.62f3 LTS, URP, Meta Avatars SDK v40.0.1, Quest 3. Parameters editable in the Inspector and an in-VR panel, no recompilation.

Takeaway. Every Meta SDK divergence traces back to a stage the SDK resolves before the hook, or does not expose to it at all.

Backup — Meta SDK: why hook injection

The SDK owns the rendering, so code can only enter at one point

Why a Hook

- Shader source is **generated** from SDK templates, not editable
- No standard URP material slots for custom shaders
- Skinning on the GPU, posed mesh in **internal buffers**
- Only entry point: `AppSpecificPostManipulation`
- Runs **per fragment**, after all SDK lighting
- Takes the lit colour, returns a modified colour

Why Use the Meta SDK at All

- Built-in body tracking: avatar driven live by the wearer
- No external motion capture needed

What the Hook Cannot Do

- No **second pass**, no screen-space post-processing
- No **depth or normal buffer**
- No avatar mask: an extra pass draws the **bind pose**
- Input is the **composited** colour, not the raw albedo

What That Changed

- V2: quantises the final RGB, not a lighting term
- V4: Sobel on lit colour, plus seam rejection
- V5: camera-distance derivative instead of depth
- V6: no blur pass, so an **unblurred** structure tensor

Backup — Why these eight techniques

Each one answers a limitation of the one before

Technique	What it adds	What it still misses	Outcome
V1 Geometric Silhouette Outline	silhouette from geometry	no interior detail	**kept**: base outline
V2 Toon Shading	quantised shading bands	no control over abstraction level	replaced by V2.2
V2.2 X-Toon Shading	2D ramp: lighting and abstraction	shading only, no edges	**kept**: manual mode
V3 Screen-Space Normal Edge Detection	creases from normal derivatives	colour boundaries: eyebrows, lips	**excluded**: facial readability
V4 Sobel Edge Detection	colour edges	noisy on photographic texture	**kept**: after texture preprocessing
V4.2 Gaussian Prefiltered Sobel	smoothing before differentiation	no gain after preprocessing, up to 72 samples	**excluded**
V5 Hierarchical Edge Detection with Multiple Cues	depth, normal and colour cues merged	facial segments unclear, nose bridge artefact	**excluded**: parameter study, $p < .001$
V6 Kuwahara Painterly Filter	painterly surface	features lost at a painterly radius	**excluded**: parameter study, $p = .003$

Order: shape, then shading, then edges, then post-processing. Final composite: V1 + V2.2 + V4.

Backup — The four assessment criteria

Unity editor and Quest 3, idle animation, several distances including 1 m

Criterion	What it checks	Why it matters
Facial readability	eyes, mouth boundary, nose bridge and jaw line stay legible	the face carries the advisor; RQ2 asks for legibility
Temporal stability	edges stay stable while the avatar moves	stimuli are animated video; screen-space methods flicker
Artefact suppression	no seam lines, hair card outlines, z-fighting or false skin edges	an artefact reads as a rendering error
Expressive abstraction	the style reads as a deliberate choice	RQ2 asks for abstraction, not noise

Facial readability separated the techniques most: it failed for V3, V4.2, V5 and V6.

Backup — V1 — Geometric Silhouette Outline

Inverted hull, drawn from geometry alone

Mixamo (Jody)

Meta SDK

Avaturn

Backup — V2 — Toon Shading

The diffuse term quantised into bands

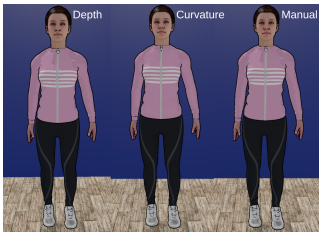
Mixamo (Jody)

Meta SDK

Avaturn

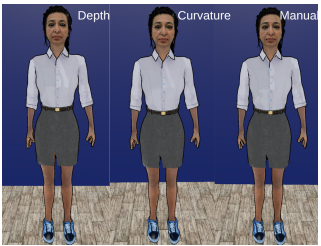
Backup — V2.2 — X-Toon Shading

A 2D ramp: lighting on one axis, abstraction on the other



Mixamo (Jody)

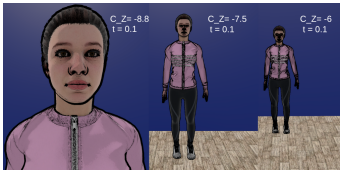
Meta SDK



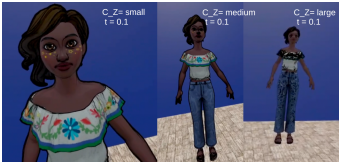
Avaturn

Backup — V3 — Screen-Space Normal Edge Detection

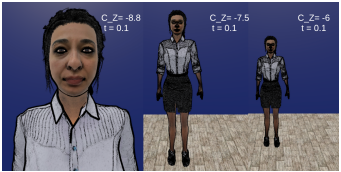
Normal derivatives. Creases yes, colour boundaries no



Mixamo (Jody)



Meta SDK



Avaturn

Backup — V4 — Sobel Edge Detection

Gradient on the colour buffer, at rising thresholds

Mixamo (Jody)

Meta SDK

Avaturn

Backup — V4.2 — Gaussian Prefiltered Sobel

Smoothing before differentiation

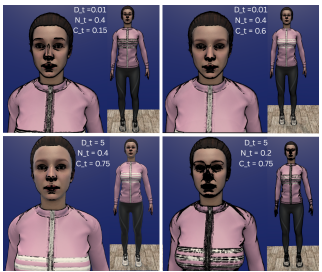
Mixamo (Jody)

Meta SDK

Avaturn

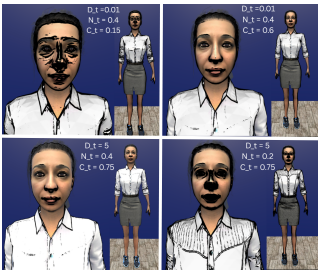
Backup — V5 — Hierarchical Edge Detection with Multiple Cues

Depth, normal and colour layers merged



Mixamo (Jody)

Meta SDK



Avaturn

Backup — V6 — Kuwahara Painterly Filter

Anisotropic painterly filtering of the surface

Mixamo (Jody)

Meta SDK

Avatum

Backup — V1 and V2 per platform

Technical review 2/6: built, input and result

	Mixamo and Avaturn (shared code, open URP)	**Meta SDK** (one fragment hook, after SDK lighting)
V1 Geometric Silhouette Outline		
Built	Outline pass: front faces culled, vertices pushed out along the normal, $\mathbf{P}' = \mathbf{P} + \mathbf{N}w$, flat colour. Depth **written**, standard test, depth bias against z-fighting	Same displacement and culling. Depth write **disabled**, looser depth test, to avoid clashing with the SDK's depth compositing
Input	Alpha cutout 0.1 clipped in **both** passes, so hair cards and eyelashes keep their shape	Transparency already **resolved by the SDK**, no manual cutout
Result	Same stable silhouette on all three, vertex stage only. Limit: outer contour only, flat planes give no outline	
V2 Toon Shading		
Built	Lambert $\mathbf{N} \cdot \mathbf{L}$ quantised, $\lfloor s \cdot n \rfloor / n$, **4 bands**, a narrow smoothstep per band edge, shadow strength remap, ambient and rim highlight	**Posterises the lit RGB** per channel, rounding to the nearest band (floor would darken), then a saturation control
Input	The lighting term, before shading	The finished SDK colour, already containing subsurface scattering, hair shading and eye glints
Result	Same flat-band look on all three. Limit: bands tied to the lighting, no interior edges	

Backup — V2.2 and V3 per platform

Technical review 3/6: built, input and result

	Mixamo and Avaturn (shared code, open URP)	**Meta SDK** (one fragment hook, after SDK lighting)
V2.2 X-Toon Shading		
Built	2D ramp: U = remapped **N** · **L**, V from **depth, curvature or manual** mode. Abstraction compression flattens bands, shadow colour blend, Blinn-Phong highlight	Same ramp. World position passed in as an **extra argument**, light and shadow fetched per fragment
Input	Normal field abstraction with a **blurred normal map** through the tangent frame. Shadow coordinate per vertex	**No tangent frame** in the hook: geometric smoothing only. Depth mode equals manual, head-and-shoulders depth range too narrow
Result	At the 1 m study framing depth and curvature hardly vary, so **manual mode** was used, detail bias 0.5	
V3 Screen-Space Normal Edge Detection		
Built	Identical everywhere: `ddx`/`ddy` of the world normal, magnitude M_N, smoothstep around threshold $\tau \pm \sigma$, blended to the line colour	
Input	Normal map decoded and rotated by the tangent frame, **one extra sample**	SDK's composited normal, **no extra sample**, but only interpolated geometric detail
Result	Rich edge network on clothing folds and seams	**Much sparser** interior lines
	All three: line width grows with distance, colour boundaries (brows, lips) missed	

Backup — V4 and V4.2 per platform

Technical review 4/6: built, input and result

	Mixamo and Avaturn (shared code, open URP)	**Meta SDK** (one fragment hook, after SDK lighting)
V4 Sobel Edge Detection		
Built	8 neighbour taps at UV offsets, BT.709 luminance, G_x, G_y, magnitude against **one threshold**	Same gradient, plus a **band gate** (min and max) and **seam rejection** on the luminance range
Input	**Raw albedo**: material boundaries, not lighting	**Composited PBR colour**: shadows, specular and UV seams also respond
Result	Mixamo clean at $t = 0.2$. Avaturn: **baked lighting** at the nose bridge, $t = 0.3$ removes facial contours. Fixed by texture preprocessing	Lower threshold needed, edges cleaner than Avaturn, less photographic noise in the input
V4.2 Gaussian Prefiltered Sobel		
Built	Shared helper: each of the 8 taps a 9-tap weighted blur, weights normalised, **up to 72 samples**. Band gate replaces the seam gate, four smoothsteps and a power curve sharpen the result	
Controls	Four sharpening steps toggled separately. Presets Light, Moderate, Aggressive (default Light)	One **tightness scalar** drives all four steps
Result	Mixamo: fewer compression artefacts. Avaturn: **baked illumination remains**, presets amplify it	Fewer high-frequency artefacts, cleaner clothing edges

Backup — V5 per platform

Technical review 5/6: hierarchical edges after AHEAD, three cues fused by max-pooling

	Mixamo and Avaturn (shared code, open URP)	**Meta SDK** (one fragment hook, after SDK lighting)
Architecture	**URP renderer feature**, a full-screen post-processing pass	**Per-fragment hook**, no extra pass
Depth layer	Roberts Cross on the **scene depth texture**, linear eye depth, normalised	`ddx`/`ddy` of the **distance to the camera**, normalised. Cannot see the background
Normal layer	Roberts Cross on the scene normal buffer. Replaces AHEAD's gradient structure layer, since baked light would read as geometry	V3 normal derivative plus the **AHEAD dark-area gain**, $\min(1/(3L + 0.1),\ 4)$
Colour layer	Roberts Cross on luminance plus a **chromatic term** for equal-brightness edges	**XDoG**: two blurs subtracted, tanh threshold, cancels broad baked lighting
Gain and fusion	Brightness-based factor, full on depth and colour, half on normal. Max-pooling	Max-pooling, then one extra smoothstep
Isolation	**Avatar mask** pass on a dedicated rendering layer	Not needed, the hook only runs on avatar fragments
Result	Richer interior lines, but **no clear facial segments**, false lines at the nose bridge	**No outer silhouette**, XDoG reduces false responses

Excluded by the parameter adjustment study, R5, $p < .001$. Mixamo and Avaturn source: `HierarchicalEdgeDetection.shader`.

Backup — V6 per platform

Technical review 6/6: anisotropic Kuwahara after Kyprianidis

	Mixamo and Avaturn (shared code, open URP)	**Meta SDK** (one fragment hook, after SDK lighting)
Passes	**Three**: (1) structure tensor from a Sobel gradient, (2) 5-tap separable Gaussian blur of the tensor, (3) the filter	**One**, inside the hook
Tensor	Sobel on luminance, **blurred** before use	`ddx`/`ddy` of the lit luminance, **unblurred**, carries lighting too
Ellipse	Orientation and anisotropy from the eigenvalues, clamped linear blend of the axes	Same eigen decomposition, Kyprianidis ratio formula with an extra eccentricity parameter
Sampling	Every pixel in the radius sorted into **8 sectors**, cost grows with the radius	Fixed **3 samples per sector** plus one shared, constant cost
Weight	$1/(1+\sigma^q)$, the 2009 form, variance passed in, exponent halved	$\max(\tau,\sigma)^{-q}$, the thresholded 2011 form
Composite	Through the avatar mask, as a post-process	Blended by a strength parameter
Result	Coherent painterly regions, **strokes follow** clothing and facial contours	**No coherent regions**, local colour saturation amplified

All three: the painterly look only appears at a large radius, where brows, lips and irises disappear. Excluded by the parameter adjustment study, R4, $p = .003$.

Backup — Statistical tests: rankings

People ranked pictures, so the tests work on order only, not on averages

1. Borda: who wins?	2. Friedman: random or not?	3. Wilcoxon: #1 really ahead?
Rank 1 earns the most points, last place 1 point, summed over everyone. A **score**, not a test.	Were some pictures **consistently** ranked higher? If not, people ranked at random.	Did most people put the winner **above the runner-up**? If not, the top two are tied.

	What changed	1. Borda winner	2. Friedman	3. Wilcoxon	Decision
R1	Sobel inner line threshold	0.03 (160 vs 149 points)	**not random**, $p < .001$	tie with 0.05, $p = .31$	**0.05**, by appeal question
R2	Outline width, depth offset	0.003, offset on	**not random**, $p < .001$	tie with 0.005, $p = .82$	**0.003, on**, Borda winner kept
R3	Shadow strength, detail bias	0.5 / 1.0	**random**, $p = .53$	—	**0.0** / **0.5**, by appeal question
R4	Kuwahara hardness, sharpness	**none**	**not random**, $p < .001$	**real win**, $p = .003$	Kuwahara **dropped**
R5	Hierarchical depth threshold	**none**	**not random**, $p < .001$	**real win**, $p < .001$	Hierarchical layer **dropped**

45 participants, 40 to 44 valid ballots per round. Tests after Demšar [24]. One Wilcoxon per round, no correction. Main study: same tests, Holm (slide 26).

Backup — Design compared with Canales

Canales et al. 2024 [7]: what was adopted, and what was changed

	Canales et al. 2024 [7]	This study
Adopted		
Trust model	Mayer et al.: ability, benevolence, integrity	same
Rating scale	7-point Likert	same
Behaviour	choose one doctor for a second opinion, and the least preferred	**rank** all three advisors: consult, trustworthy, unsettling
Propensity	7-item propensity to trust survey	single item: *In general, I tend to trust people I meet for the first time*
Changed		
Medium	immersive VR, Meta Quest 2, interactive interview	prerecorded video in a browser, no interaction
Identity	three doctors, identity crossed with style	**one speaker**, identity held constant
Conditions	caricature, mid, realistic	real video, photorealistic avatar, NPR avatar
Scenario	one medical scenario, kidney lump, second opinion	three everyday decisions: career, travel, learning
Speech	answers to three questions, one per trust dimension	one recommendation, about 22 s
Trust items	5 ability, 5 benevolence, 6 integrity, 4 overall trust	3 + 3 + 3, no overall trust items
Order	six identity and style combinations	3×3 Latin square, three versions
Analysis	linear mixed models, exact multinomial test	RM-ANOVA with Holm, TOST, multilevel regression, Friedman and Wilcoxon
Sample	$n = 66$	$n = 41$

Backup — The three scenarios

Order held constant, conditions rotated by version

	Scenario	The two options	Advisor picks	Written with in mind
Video 1	Career choice	Job A generalist, lower pay Job B specialist, higher pay	**Job A**, presented first	Ability
Video 2	Travel route	Route A direct, 10 min spare Route B one transfer, 40 min spare	**Route B**, presented second	Benevolence, Ability
Video 3	Learning resource	Option A course, EUR 350, certificate Option B free videos, no certificate	**Option A**, presented first	Integrity

Counterbalancing

Version	Video 1	Video 2	Video 3
A	C2	C1	C3
B	C3	C2	C1
C	C1	C3	C2

Design Rules

- **Everyday decisions** for a 20 to 35 year old student sample
- Equal detail, **one clear cost** per option
- Recommendation alternates **A, B, A**, not always presented last
- Same **five-step script** every time, see the vignette slides
- Baseline recorded **before** the advisor speaks

Backup — Vignette 1, Career choice

Shown on screen before the video

You recently completed your master's degree and have received two job offers. Both start in one month. You have five days to decide.

Job A is a generalist role at a smaller company. You would define your own work area, your contributions would be directly visible to the team, and the skills you build would transfer across industries. The starting salary is below the industry average for your field.

Job B is a specialist role at a large, well-known company. The employer name carries weight on a CV and the role is in a high-demand field. However, the position involves deep specialisation within a large team, with limited individual visibility. The starting salary is above the industry average for your field.

Advisor, spoken in the video

(1) *"I'd go with Job A.* (2) *Yes, the salary is lower right now,* (3) *but the generalist role gives you a range of experience that is genuinely hard to get later. Salary gaps can be closed in the future.* (4) *But once you specialise, switching back is costly.* (5) *Take the pivot while you can."*

(1) recommendation (2) its drawback (3) its key advantage, after a contrastive *but* (4) weakness of the alternative (5) restatement

Backup — Vignette 2, Travel route

Shown on screen before the video

You have just arrived by train in a city you have never visited. You need to reach your hotel, where colleagues are expecting you in one hour.

Route A is a direct route on a single line with no transfers. It is straightforward to navigate in an unfamiliar city. If the journey runs on time, you would arrive approximately 10 minutes before your meeting begins.

Route B involves one transfer at a mid-size station you have not used before. If the connection runs on time, you would arrive approximately 40 minutes before your meeting begins. However, the transfer station is unfamiliar and a missed connection could leave you with very little time.

Advisor, spoken in the video

(1) *"I'd take Route B.* (2) *Yes, it has a transfer at a station you have never used.* (3) *But it gets you there forty minutes early. If something goes wrong, you still have enough time to recover.* (4) *Route A is direct but leaves only ten minutes. That may not be enough.* (5) *Take the buffer."*

(1) recommendation (2) its drawback (3) its key advantage, after a contrastive *but* (4) weakness of the alternative (5) restatement

Backup — Vignette 3, Learning resource

Shown on screen before the video

You want to develop a professional skill that would strengthen your CV and improve your career prospects. You have a few weeks available and you are a student with a limited budget.

Option A is a structured online course from a well-known learning platform. It follows a clear curriculum and awards a recognised certificate you can add to your CV. It costs EUR 350.

Option B is a video tutorial series by a practitioner with a strong reputation. It covers the same material at no cost and you set your own pace. However, there is no certificate and no structured curriculum.

Advisor, spoken in the video

1 *"I would take Option A.* 2 *Yes, 350 euros is a real cost.* 3 *But the certificate is what matters here. It is not just about what you learn, it is about what you can show.* 4 *A recruiter cannot verify a free tutorial series.* 5 *That is worth it."*

1 recommendation 2 its drawback 3 its key advantage, after a contrastive *but* 4 weakness of the alternative 5 restatement

Backup — Hypotheses and their tests

Each prediction follows a prior finding

Hypothesis	Grounded in	Test	Why this test
H1: C3 less eerie than C2	Dubosc 2025 [8]: stylised avatars rated less eerie	RM-ANOVA, Holm pairwise	one subscale, three related conditions
H2: C3 no lower trust than C2	Canales 2024 [7]: style did not lower self-reported trust	pairwise + TOST, $d = 0.3$	a null needs an equivalence test
H3: eeriness lowers, likeability raises trust	Ho 2010 [3]: eeriness distinct from dislike; Alipour 2026 [21]: link not uniform	person-centred multilevel regression	separates within-person change from general trustingness
H4: C3 no lower trust than C1	Gao 2025 [9]: real speaker above the stylised agent	pairwise + TOST, $d = 0.3$	same as H2, against the real speaker

H2 and H4 predict *no difference*. A non-significant p is not evidence of equality, so both carry a TOST.

Backup — The rating instruments, and why each

Canales [7] supplied the trust battery only, so the remaining scales come from elsewhere

Part	Source	Format	Why this one
B Likeability	**Godspeed**, Bartneck et al. 2009 [14]	5 items, semantic differential 1–5	Canales measured no likeability. Zell [15] shows appeal moves with material while realism does not
D Eeriness	Ho & MacDorman 2010 [3]	8 items, semantic differential 1–5	Built to separate **eeriness from human likeness**, which the Godspeed indices confound. H1 needs that
E–G Trust	Mayer et al. 1995 [6], wording after Canales [7]	3+3+3, Likert 1–7	Three separable dimensions, so no single trust score is ever reported
H Stylisation	own item	1 item, Likert 1–7	Manipulation check: verifies the difference was **seen**, so a null is not an unnoticed manipulation

Why Godspeed, not an ad hoc scale

- Published reliability and validity, and widely reused, so results stay comparable
- One response format shared by B and D

Why no human-likeness scale

- Godspeed's anthropomorphism subscale was **dropped** before the study ran, and it **confounds** with the eeriness items
- Human likeness is a **concept** here, never a measured variable

What the form calls them

- Titles are **neutral** so nobody is primed by a construct name
- *Warmth and Appeal* = B, *Impression* = D, *Trust* = E–G, *Overall Impressions* = rankings

Backup — Trust items: Ability and Benevolence

Canales et al. 2024 [7] next to this study, word for word

This study	Closest Canales et al. 2024 item	What changed
Ability		
This advisor is capable and competent.	A1: The doctor is very capable of performing his/her job.	doctor → advisor, no job reference, *competent* added
This advisor seems knowledgeable.	A2: The doctor has much knowledge about the work that needs done.	shortened, no task reference
This advisor seems to have the expertise needed to give sound advice on this kind of decision.	A4: The doctor has specialized capabilities that can increase performance. A5: The doctor is well qualified.	tied to the advice decision, not medical performance
Benevolence		
This advisor will act in a benevolent (well meaning) manner.	Pretest prompt: This person will behave in a benevolent (well meaning) manner.	*behave* → *act*
This advisor would act in my best interest.	B4: The doctor really looks out for what is important to me.	rephrased after the benevolence definition, *best interest of the trustor*
This advisor appears to have good intentions.	B1: The doctor is very concerned about my welfare.	**no direct counterpart**, closest item shown

Canales items from their Supplement 11.1 (post-doctor surveys). Pretest prompts from their Section 3.2, used to rate the doctors' dialogue.

Backup — Trust items: Integrity

And what was left out of Canales et al. 2024 [7]

This study	Closest Canales et al. 2024 item	What changed
Integrity		
This advisor adheres to a set of principles that I find acceptable.	Pretest prompt: The person adheres to a set of principles that I find acceptable. I6: Sound principles seem to guide the doctor's behavior.	subject only
This advisor appears to be honest.	**no direct counterpart**	—
This advisor seems sincere.	**no direct counterpart**	—

Not Carried Over

- A3: I feel very confident about the doctor's skills
- B2, B3, B5: needs and desires, not hurt me, go out of his/her way
- I2 to I5: sense of justice, stick to word, fair dealings, values
- All four **overall trust** items, two of them reverse scored
- I1 is **blank** in the published supplement

Kept the Same

- Mayer et al. 1995 [6]: ability, benevolence, integrity
- 7-point Likert, agree to disagree
- Rated once per advisor, after each advisor
- Three dimensions analysed separately

Backup — Composite reliability

Chapter 6.2.3 — Cronbach's α per subscale

Subscale	**Items** k	α
Likeability	5	.94
Eeriness	8	.87
Ability	3	.96
Benevolence	3	.95
Integrity	3	.92

Every subscale clears the conventional .70 threshold, and each one was recomputed separately within C1, C2 and C3 rather than on the pooled data alone. All five held in every condition.

Averaging the items of each subscale into a single composite is therefore justified, and every test reported in this talk runs on those five composites.

Backup — Statistical analysis, step by step

The order in which the main study was analysed, $n = 41$

	Method	Question it answers	Result here
1	**Cronbach's** α, overall and per condition	Do the items of one subscale measure one thing? Minimum .70 [3,14]	.87 to .96 overall, never below .80 per condition. Composites formed
2	**One-way RM-ANOVA**, one per subscale	Does condition change the rating at all? Same people in all three	All five subscales significant, $p < .001$
3	**Mauchly**, **Greenhouse-Geisser**	Do the pairwise differences vary about equally? Correct the test if not	Correction applied where violated, e.g. stylisation question, $W = .53$
4	**Holm** pairwise, η_p^2, Hedges' g	Which pairs differ, and by how much?	C1 differs from both avatars on all five. **C2 vs C3 on none**
5	**TOST**, $d = 0.3$ [23]	Are two conditions practically equal? H2, H4	H2 inconclusive, $p = .240$. H4 not equivalent, $p = .998$
6	**Person-centred multilevel** regression	Within a person, do eeriness and likeability predict trust? H3	All slopes $p < .001$, in the predicted directions
7	**Friedman**, Holm **Wilcoxon**	Do the three rankings differ from chance? Which pairs?	C1 first on consulting and trust. C2 and C3 tied
8	ANOVA on **position** and **study version**	Did the order of the videos matter?	Position: no effect. Version: likeability and eeriness

Takeaway. Ratings are parametric, one subscale at a time. Ranks are non-parametric. A null prediction gets an equivalence test.

Backup — Statistical tests: ratings

Main study, five subscales, three conditions

Test	What it answered	Param. study	Main study	Source or precedent
Power analysis	Minimum sample: $n = 38$ at $\eta_p^2 = .06$, power .90	—	✓	—
Cronbach's α	Do the items of a subscale measure one thing? ($\geq .70$)	—	✓	Bartneck 2009 [14], Ho 2010 [3]
Repeated-measures ANOVA	Effect of condition, one subscale at a time	—	✓	Zell 2015 [15], McDonnell 2012 [19], Latoschik 2017 [22]
Mauchly, Greenhouse-Geisser	Sphericity check, and correction where violated	—	✓	—
Holm pairwise, η_p^2, Hedges' g	Which conditions differ, and by how much	—	✓	—
TOST, $d = 0.3$	Are two conditions equivalent? H2 and H4	—	✓	Lakens 2017 [23]
Multilevel regression	H3: eeriness and likeability predicting trust within each person	—	✓	—
ANOVA on position and version	Order checks: video slot, and study version group	—	✓	—
Pearson correlation	Does trust propensity predict the trust composite?	—	✓	Mayer 1995 [6], same check in Canales 2024 [7]

Backup — Power analysis

How many participants the study needed, and for which test

What Power Means

Power is the chance that the study **detects an effect**, if the effect is real. Fix the test, the effect size, α and the power, and the analysis returns the sample size.

Input	Value
Test	RM-ANOVA, 3 conditions, within subjects
Effect size	$\eta_p^2 = 0.06$, medium
α	.05
Power	0.90
Minimum	$n = 38$

Difference versus Equivalence

Question	Needed	At $n = 41$
Is there a medium difference? RM-ANOVA, H1	$n = 38$	**covered**
Are two conditions equal? paired TOST, $d = 0.3$, H2 and H4	$n \approx 120$	**power** ≈ 0.20

Recruitment

Target **48**, 16 per study version, to allow for exclusions. 52 took part, 11 excluded, **41 valid**. The target was set by the ANOVA, which the primary hypotheses depend on.

Takeaway. Planned to detect a medium difference, not to prove equality. C2 vs C3 came out small ($g \approx 0.07$ to 0.20), so: no difference *detected*, equivalence inconclusive.

Backup — Why the bound is $d = 0.3$

The bound is the smallest trust difference that would matter, fixed before the data.

- The bound says **how small** a difference counts as *no difference*, and is set **in advance** [23]
- $d = 0.3$ follows **Lakens's own example** of a smallest effect of interest [23]
- Thesis: the smallest difference in trust treated as **practically meaningful**

± 0.51

trust points, C2 vs C3: $0.3\times$ the average SD

± 0.47

trust points, C1 vs C3: about half a scale point

$d = 0.2$

stricter, but needs about **2.25**$\times$ the people, well over 250

$d = 0.3$

chosen: meaningful, about 120 people for power .90

$d = 0.5$

too generous: would call about **0.85 points** *equal*

Backup — Why a multilevel model for H3

H3 is about what happens inside one person, so a simple correlation would mix two things.

The problem with a simple correlation

- 41 people × 3 advisors = **123 ratings**, three from each person
- A pooled correlation mixes *"I trust **this advisor** less"* with *"I am a **less trusting person**"*
- Ratings from one person are **not independent**

What the model does

- **Person-centring:** each rating minus that person's own mean
- **Random intercept:** every person keeps their own baseline trust

123

ratings from 41 people

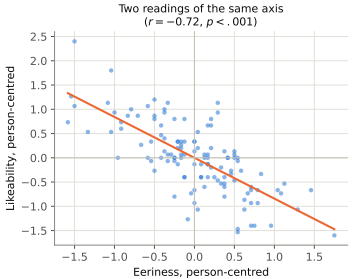
.48 to .63

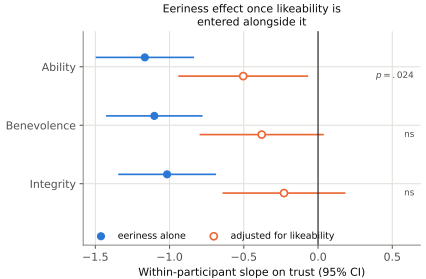
intraclass correlation: half or more of the trust variance is **between** people

That share is why the random intercept was needed.
Results on slide 24.

Backup — Eeriness against likeability

The two move together ($r = -.72$), so I cannot say which one drives trust.





52%
shared variance

−0.50
eeriness on ability, joint model

3 of 3
trust subscales where likeability holds

Joint model: eeriness holds on ability ($p = .024$), drops out on benevolence ($p = .075$) and integrity ($p = .273$).

Backup — Why a ranking, and how it was read

A ranking of all three, because a single choice would all go to the real speaker.

Why rank, not pick one

- C1 was expected to win, so a single pick would **pile onto C1**
- Then nothing would be left to separate **C2 from C3**
- Adapted from the single-choice task of Canales [7]

What it measures

- Ratings: what people **believe** about the advisor
- *Most want to consult*: whether they would **act** on it, the primary behavioural measure

3

rankings: consult, trustworthy, unsettling

Friedman

then Holm-corrected Wilcoxon on the three pairs.

Ranks are ordinal

Unsettling: the question read *most to least*, but the columns said 1 (Best) to 3 (Worst). Null either way, $p = .60$.

Backup — Full statistics table

The real speaker differs from both avatars on every subscale. The avatars differ on none.

Subscale	Omnibus	C1 vs C2	C1 vs C3	C2 vs C3
Likeability	$F(2, 80) = 26.85$, $\eta_p^2 = .40$	$g = 0.92$	$g = 1.02$	$p = .54$, $g = 0.08$
Eeriness	$F(2, 80) = 20.39$, $\eta_p^2 = .34$	$g = 0.98$	$g = 0.98$	$p = .66$, $g = 0.07$
Ability	$F(2, 80) = 10.91$, $\eta_p^2 = .21$	$g = 0.53$	$g = 0.75$	$p = .21$, $g = 0.20$
Benevolence	$F(2, 80) = 14.31$, $\eta_p^2 = .26$	$g = 0.59$	$g = 0.79$	$p = .18$, $g = 0.19$
Integrity	$F(2, 80) = 12.47$, $\eta_p^2 = .24$	$g = 0.66$	$g = 0.80$	$p = .27$, $g = 0.16$

All omnibus $p < .001$. All C1 comparisons significant: $p = .004$ and $p = .001$ on C1 vs C2 ability and integrity, otherwise $p < .001$.

Order checks

- Video position: **no effect** on any subscale
- Study version: likeability ($p = .002$) and eeriness ($p = .008$), **no trust subscale**

H3 models

- ICC: eeriness .48 to .57, likeability .57 to .63
- Half or more of trust variance is **between** people

Backup — The mismatch evidence

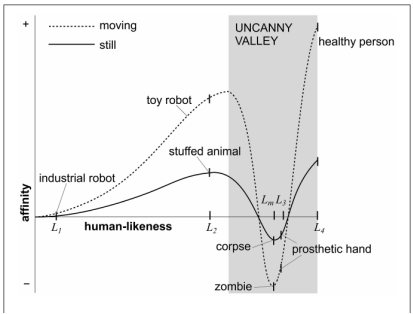
Eeriness comes from a mismatch between features, not from realism itself.

Kätsyri et al. 2015 [17]

- Review of **17 studies**, testing the competing valley explanations
- *More human-like means a dip*: held in only **1 of 8**
- **Perceptual mismatch**, features at different realism levels: **4 of 4**
- One study: a **mismatched face and voice** was rated eerier

Cihodaru-Ștefanache & Podina 2025 [16]

- Recent review of embodied conversational agents
- Reaches the **same conclusion**: mismatch, not realism



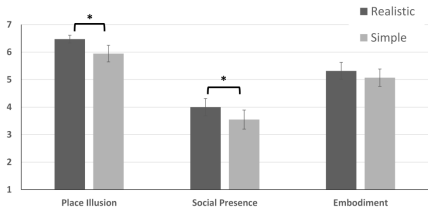
Kätsyri et al. [17]

Link to C3: a human voice and human motion on an abstracted face is exactly such a mismatch.

Backup — Design compared with Zibrek

Same manipulation as Zibrek & McDonnell [18], but on a flat screen instead of in VR.

Zibrek & McDonnell [18]: realistic and simple



Their result [18]

Same

- Only render style changes
- One recorded performance
- Self-report plus behaviour

Different

- VR body vs **flat video**
- Room stylised too
- Presence, not trust

$d = 0.56$

realism raised social presence. Behaviour did not change

$n = 19$ in VR, against $n = 41$ here

Backup — Counterbalancing options

Three versions was what 41 participants allowed.

3	**6**	**36**
versions, Latin square, about 14 people each. **Used**	versions, all condition orders, about 7 people each	versions, every condition and scenario order ($3! \times 3!$), about 1 each

- The Latin square puts each condition **once in each position**
- Scenario order was held fixed, so only the conditions rotated
- Six versions would need about **twice the sample**
- **Cost:** carryover between specific pairs, such as C1 $\rightarrow$ C3, cannot be estimated

Backup — Which version differed, and on what

The groups differed on eeriness and likeability, not on trust.

C1 sat	Ver.	n	Eerie	Liked
1st	C	14	**3.01**	**2.91**
2nd	A	15	2.76	3.33
3rd	B	12	**2.32**	**4.04**

- Likeability $p = .002$, eeriness $p = .008$
- Trust: **all null**, $p = .14$ to $.30$

- The **later the real speaker** came, the milder the ratings
- Versions B and C carry it: $g = 1.50$ likeability, $g = -1.11$ eeriness
- Fits an **anchoring** effect: a real human first makes what follows look worse
- **H1 survives:** C2 minus C3 eeriness does not follow it, $p = .244$
- **Cannot rule out:** a version is a group *and* an order

Supplementary analysis, not in the thesis. Likeability's C2 minus C3 contrast does vary, $F = 3.43$, $p = .043$, uncorrected.

Backup — Which condition moved

Both avatars moved together, so the contrast H1 tests was preserved.

Eeriness, 1–5

C1 sat	C1	C2	C3
1st (C)	2.47	**3.37**	3.18
2nd (A)	2.12	2.92	3.23
3rd (B)	2.01	**2.46**	**2.48**
p	.24	**.005**	**.039**

- C1 **holds still**
- Both avatars move **together**

Likeability, 1–5

C1 sat	C1	C2	C3
1st (C)	3.61	**2.36**	2.77
2nd (A)	4.04	3.23	2.73
3rd (B)	**4.53**	**3.90**	**3.70**
p	**.009**	**.001**	.050

- Even C1 shifts
- Strongest on **C2**

Trust composite, 1–7

C1 sat	C1	C2	C3
1st (C)	4.94	3.48	3.75
2nd (A)	5.28	4.73	3.73
3rd (B)	5.64	4.70	4.50
p	.46	.08	.45

- **Nothing significant**
- H2 to H4 untouched

C2 minus C3 eeriness by version: -0.31, -0.02, $+0.19$, $F(2, 38) = 1.46$, $p = .244$. Likeability contrast varies ($p = .043$), trust contrast does not ($p = .091$). Supplementary analysis, not in the thesis.